WEEK 11 OF 15 · STAGE 3: ADVANCED METHODS

Optimisation models

One goal, many limits: how planners' models choose

New R skill lpSolve: objective, constraints, reading shadow prices

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	Week 4 Economics primer	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	Week 8 Energy in GCAM	Week 9 Land, water and bioenergy	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	**Week 11 Optimisation IAMs**	Week 12 Why models disagree	Week 13 Uncertainty	Week 14 Frontiers and limits	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Write a small energy problem as an objective and constraints
2. Solve a three-technology power system by hand and with lpSolve
3. Read a shadow price and explain why it acts like a carbon price
4. Contrast GCAM's simulation with optimisation and CGE models

? **Big question** If one planner could choose the whole power mix with perfect knowledge, what would they build, and what would a climate cap cost?

You are the planner

Supply 100 TWh of electricity at the lowest cost, using coal, gas and solar, while emitting no more than 40 Mt of CO2.



Would you use all the solar you are allowed?

AN EVERYDAY PICTURE

An optimisation model is a sat-nav for an energy system



You give a sat-nav one goal, the fastest route, and the limits: roads, one-way streets, a closed bridge. It searches and returns the single best route. An optimisation model does the same with cost as the goal and demand, plant sizes and emission caps as the limits.

What it means for models

1. Goal = objective (lowest total cost)
2. Limits = constraints (demand, capacity, cap)
3. Answer = one best plan, not a range

SIDE BY SIDE

The toy power system from lab 11

Technology	Cost $/MWh	tCO2 per MWh	Limit TWh
Coal	60	0.9	60
Gas	70	0.4	50
Solar	80	0	30
Demand			100 in total



Same costs and emission factors as the week 4 market. Illustrative numbers, not a real country.

Worked example: no cap, fill the cheapest first

Problem Meet 100 TWh at the lowest cost with no emissions cap.

1. Coal is cheapest: use its full limit, 60 TWh
2. Still need 100 − 60 = 40 TWh: take it from gas, the next cheapest
3. Cost = 60 × 60 + 40 × 70 = 3,600 + 2,800 = $6,400 million
4. Emissions = 0.9 × 60 + 0.4 × 40 = 54 + 16 = 70 Mt

Answer

Coal 60, gas 40, solar 0 TWh; $6,400 million; 70 Mt CO2

Worked example: add a 40 Mt cap

Problem Same system, but emissions must be 40 Mt or less. Find the cheapest mix.

1. Swapping coal for gas costs 10 $/MWh and saves 0.5 t: $20 per tonne
2. Swapping coal for solar costs 20 $/MWh and saves 0.9 t: $22.2 per tonne
3. Do the cheaper swap first: gas to its limit, 50 TWh, using 0.4 × 50 = 20 Mt
4. That leaves 40 − 20 = 20 Mt for coal: 20 ÷ 0.9 = 22.2 TWh
5. Solar fills the rest: 100 − 50 − 22.2 = 27.8 TWh
6. Cost = 22.2 × 60 + 50 × 70 + 27.8 × 80 ≈ $7,056 million

Answer **Coal 22.2, gas 50, solar 27.8 TWh; about $7,056 million**

NUMBERS TO REMEMBER

The solver gives you a carbon price for free

$22.2

per tonne: the shadow price of the 40 Mt cap

+$656m

extra cost of the cap: 7,056 − 6,400

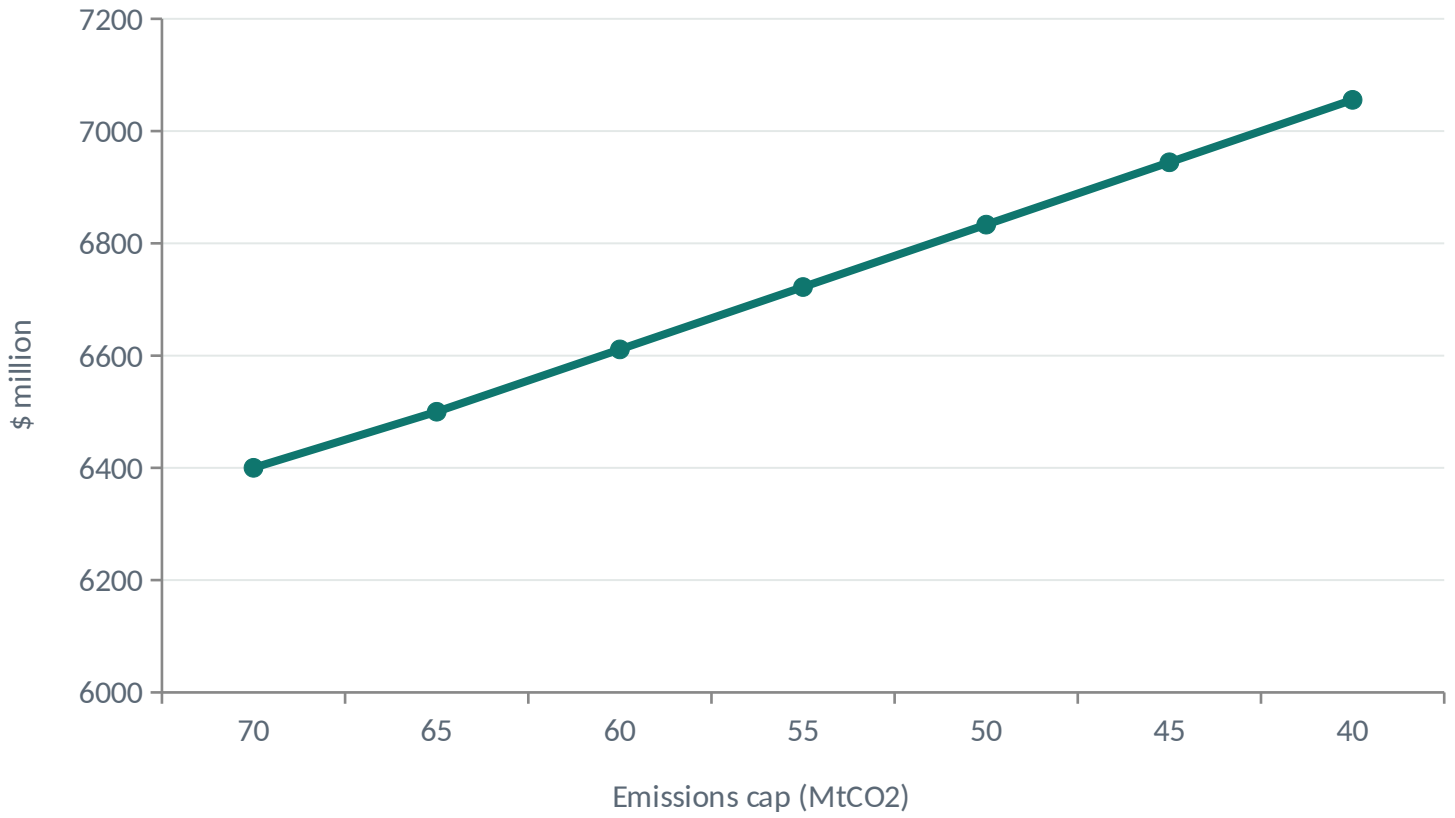
38 Mt

lowest cap with any solution at all

Shadow price = (80 − 60) ÷ 0.9: the cost gap between solar and coal per tonne saved. Illustrative.

PICTURE THE NUMBERS

Tighter caps cost more, at a rising price per tonne



Computed by solving the lab 11 system at each cap. Illustrative numbers.

Read it like this

1. 70 to 65 Mt: coal swaps for gas at $20 per tonne
2. Below 65 Mt: coal swaps for solar at $22.2 per tonne
3. The slope of this line is the shadow price

Quick check

Why does a 37 Mt cap make lpSolve report status 2, no solution?

A The solver ran out of time

B Even the cleanest mix allowed emits 38 Mt

C Solar is too expensive

D Demand is not a constraint

Quick check: answer

Why does a 37 Mt cap make lpSolve report status 2, no solution?

A. The solver ran out of time

B. Even the cleanest mix allowed emits 38 Mt ✓

C. Solar is too expensive

D. Demand is not a constraint

COMPARE

GCAM simulates; optimisation models plan

Simulation: GCAM

- Shares the market across options by relative cost (logit)
- Decides one period at a time with no view ahead (myopic)
- Asks: what might happen if a cost or policy changes?

Optimisation: MESSAGEix, REMIND, TIMES

- Picks the single least-cost plan
- Usually sees every period at once (perfect foresight)
- Asks: what is the best plan, and what does a target cost?

KEY IDEA

How far does the model look across the economy?

1

Partial equilibrium

Balances some markets only. GCAM balances energy, land and water markets and stops at the energy bill.

2

General equilibrium (CGE)

A computable general equilibrium model follows every sector, wages, incomes and trade, so it can ask who gains and who loses.

3

The trade-off

A CGE model sees the whole economy but with little technology detail: power may be one sector with a few options.

WATCH OUT

Common mix-ups

Not quite	Better
Optimisation models show what will happen	They show the cheapest plan for a planner who knows the future perfectly. Nobody does.
The shadow price is −22.2, so the cap makes money	The minus sign only says a tighter cap raises cost. Read the size, 22.2, as the carbon price.
The solver tries every possible mix	It moves along the corners of the allowed region and stops when no neighbour is cheaper.

NEW WORDS THIS WEEK

Words to add to the class glossary

Linear program A problem with one goal to make as small (or large) as possible and limits written as simple sums, solved by a computer.

Objective The single number the model tries to make as small as possible, here the total cost of electricity.

Constraint A rule the answer must obey, such as meeting demand, staying under a plant's limit, or staying under an emissions cap.

Shadow price How much the objective would change if one constraint were loosened by one unit. For an emissions cap it is a carbon price.

Perfect foresight The model knows every future period when it decides today. GCAM does not; it is myopic, deciding one period at a time.

HANDS-ON LAB

Lab 11: a power system as a linear program

Write the toy power system for lpSolve, find the cheapest mix with and without a 40 Mt cap, and read the shadow prices. Then watch a MESSAGEix Westeros demonstration.

Script: labs/lab11.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

LAB · STEP 1 OF 4

Write the objective and constraints

One cost per technology, then one row per constraint and one column per technology. Fill GAP 1 with demand.

```
library(lpSolve)
cost <- c(60, 70, 80)            # coal, gas, solar
A <- rbind(demand = c(1, 1, 1),
           coal   = c(1, 0, 0),
           gas    = c(0, 1, 0),
           solar  = c(0, 0, 1))
dir <- c("=", "<=", "<=", "<=")
rhs <- c(___, 60, 50, 30)        # GAP 1: demand in TWh
```

⌨ *Fill each ___ gap before running that line. Stuck? The solution file has the complete version.*

Solve without a cap

compute.sens = TRUE asks for the shadow prices. Read status first: 0 means success.

```
base <- lp("min", cost, A, dir, rhs,
           compute.sens = TRUE)
base$status      # 0 = success, 2 = no solution
base$solution    # TWh of coal, gas, solar
base$objval      # total cost, million $
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Add the 40 Mt cap

The cap is one more row: the emission factors. Fill GAP 2 with emis and GAP 3 with 40.

```
emis <- c(0.9, 0.4, 0)            # tCO2 per MWh
A2   <- rbind(A, co2 = ___)       # GAP 2
dir2 <- c(dir, "<=")
rhs2 <- c(rhs, ___)               # GAP 3: the cap
capped <- lp("min", cost, A2, dir2, rhs2,
             compute.sens = TRUE)
capped$solution
sum(emis * capped$solution)       # equals the cap
```

Fill each ___ gap before running that line. Stuck? The solution file has the complete version.

Read and check the shadow prices

duals lists one number per constraint, then one per technology. Loosen the cap by 1 Mt to check the sign.

```
capped$duals    # 5 constraints, then 3 technologies
rhs3 <- rhs2
rhs3[5] <- rhs2[5] + 1
looser <- lp("min", cost, A2, dir2, rhs3)
looser$objval - capped$objval   # about -22.2
```



Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.

Did it work?

You should see

- Without a cap: coal 60, gas 40, solar 0; cost 6,400
- With the 40 Mt cap: coal 22.2, gas 50, solar 27.8; cost about 7,056
- capped$duals[5] is about −22.2, and loosening by 1 Mt saves about 22.2

Worksheet questions

1. Use the shadow price to estimate how much total cost falls if the cap is loosened from 40 to 50 Mt. Then solve it and compare.
2. Try a cap of 37 Mt. What is the status, and why?
3. Explain in two sentences why solar stops at 27.8 TWh instead of its limit of 30.

Take-home messages

1. An optimisation model minimises one objective subject to constraints and returns one best plan
2. With a 40 Mt cap the cheapest mix is coal 22.2, gas 50, solar 27.8 TWh
3. The shadow price of the cap, $22.2 per tonne, is the carbon price that gives the same mix
4. GCAM simulates period by period; optimisers usually plan with foresight; CGE adds the whole economy

Next: week 12

Week 12 asks why different models give different answers to the same question, and places your GCAM run among other models' results.

Practise

Interactive course, week 11: try the lp explorer and the 5-question quiz.

Readings: Huppmann et al. (2019), The MESSAGEix Integrated Assessment Model and the ix modeling platform, Environmental Modelling and Software; lpSolve package documentation; GCAM v8.2 documentation: Overview (dynamic recursive), Solution and Computation